Beyond WEIRD Legal AI: Why Concept-Based Models Outperform Token-Based Models in Non-WEIRD Jurisdictions

A Theoretical Framework for Legal AI in Contexts with Structural Formal/Informal Gaps

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ABSTRACT

Most Legal AI models exhibit a WEIRD bias: trained on formal legal texts from Western, Educated, Industrialized, Rich, Democratic societies, they implicitly assume that written law predicts legal outcomes. Drawing on Henrich's (2020) comprehensive analysis demonstrating that WEIRD populations are psychological outliers across multiple dimensions, I extend this framework to legal systems and Legal AI architectures. This assumption systematically fails in non-WEIRD contexts where the gap between formal rules and informal practices is structural, not transitional.

Using Argentina as a paradigmatic case (Rule of Law Index: 0.51, labor informality: 42%, criminal enforcement: 3%), I document how this formal/informal duality is constitutive of non-WEIRD legal systems. Token-based language models, trained almost exclusively on statutes and case law, remain blind to this gap because their tokenization-first architecture fragments sentence-level semantic relationships that encode the formal/informal tension.

I propose that concept-based architectures offer structural advantages for non-WEIRD legal AI because: (1) sentence-level embeddings preserve contextual nuance lost in word-level

tokenization, (2) abstract concept spaces can model relationships between formal rules and informal practices, and (3) cross-lingual concept transfer enables learning from enforcement studies across similar jurisdictions. Meta AI's Large Concept Models (Barrault et al., 2024), which operate in sentence representation space and demonstrate superior zero-shot cross-lingual generalization, provide empirical validation for this architectural approach.

To enable empirical validation of this framework, I introduce **LegalGapDB**: an open-source, collaboratively-built database of formal/informal legal gaps. Modeled on successful crowdsourcing projects like reCAPTCHA and Wikipedia, LegalGapDB provides structured case documentation, peer-reviewed validation, academic credit via DOI for contributors, and free access for researchers. LegalGapDB (https://adrianlerer.github.io/LegalGapDB/) is an open-source database documenting formal-informal legal gaps in non-WEIRD jurisdictions.

This theoretical framework extends my prior work on Small Concept Models (Lerer, 2025) by specifying when and why concept-based architectures outperform token-based models. I argue that the 88% of humanity living in non-WEIRD contexts requires Legal AI architectures that explicitly model structural gaps, not just formal texts.

Contribution: First theoretical framework linking social science (WEIRD bias), legal sociology (formal/informal gap), and AI architecture (concept-based vs. token-based) for non-WEIRD legal systems.

1. INTRODUCTION

1.1 The WEIRD Problem in Legal AI

The field of Legal AI has achieved remarkable progress in recent years, with models like Legal BERT, GPT-4 for legal applications, and specialized systems for contract analysis and case law prediction. However, these advances share a critical limitation: they are designed for, trained on, and evaluated against legal systems from WEIRD societies.

WEIRD—Western, Educated, Industrialized, Rich, Democratic—describes the approximately 12% of humanity that comprises the primary focus of social science research, despite being statistical outliers in most psychological and sociological measures. This bias extends deeply into Legal AI:

- **Legal BERT:** Trained on 12 million words from EU legislation, US case law, and UK statutes
- **Pile of Law:** 256GB corpus, 99% from USA, UK, Canada, Australia
- **MultiLegalPile:** 689GB, primarily Common Law jurisdictions

Implicit assumption: Legal reality is captured by written law.

This assumption holds reasonably well in WEIRD contexts with high rule of law, consistent enforcement, and small gaps between formal rules and actual practice. But for the 88% of humanity living in non-WEIRD contexts, this assumption fails systematically.

1.2 The Formal/Informal Gap in Non-WEIRD Legal Systems

Consider Argentina, a paradigmatic non-WEIRD legal system:

Legal Domain	Formal Law	Informal Practice	Gap
Labor	LCT Art. 245 (severance)	42% informality	42%
Tax	32% fiscal pressure	8% GDP evasion	25%
Criminal	Due process (CPP)	[Data pending verification]	[Data pending verification]
Corporate	Governance (L.19.550)	70% SMEs no compliance	70%

Rule of Law Index: 0.51 vs. 0.74 (USA)

This formal/informal gap is structural, not transitional. It reflects weak enforcement capacity, selective prosecution, social norms divergence, and institutional distrust. A Legal AI model trained exclusively on Argentine statutes and Supreme Court rulings will systematically overestimate formal compliance because its training corpus excludes the informal half of legal reality.

1.3 Prior Work: Small Concept Models Framework

In my previous work (Lerer, 2025), I introduced Small Concept Models as an efficient adaptation of Meta's Large Concept Models framework for domain-specialized legal applications. SCMs leverage sentence-level semantic embeddings, LoRA-based efficient fine-tuning, and edge deployment for client confidentiality.

1.4 This Paper: Why SCM > SLM in Non-WEIRD Contexts

This paper provides the theoretical foundation for when and why concept-based architectures outperform token-based models. My central thesis:

In legal systems with structural formal/informal gaps, concept-based models systematically outperform token-based models because sentence-level semantic embeddings preserve the contextual relationships that encode formal/informal duality, while tokenization fragments these relationships into isolated lexical units.

I develop this thesis through four theoretical claims:

1. **Epistemological:** Non-WEIRD legal knowledge requires dual-corpus training (formal + enforcement/sociological data)
2. **Architectural:** Token-based models fragment semantic units that express formal/informal tension
3. **Representational:** Concept spaces enable explicit modeling of gap relationships
4. **Empirical (proposed):** Concept-based models should outperform token-based models on enforcement prediction tasks

1.5 Contribution and Structure

Contributions:

- First systematic analysis of WEIRD bias in Legal AI architectures
- Theoretical framework linking formal/informal gap to model architecture
- Proposal for "Gap Comprehension" as new evaluation metric for non-WEIRD legal AI
- Research agenda for cross-jurisdictional non-WEIRD legal AI

Structure:

- Section 2: WEIRD bias in social science and Legal AI
- Section 3: Formal/informal gap in Argentina (empirical documentation)
- Section 4: Why tokenization fails (architectural analysis)
- Section 5: Why concepts succeed (theoretical advantages)
- Section 6: LegalGapDB collaborative infrastructure
- Section 7: Discussion, limitations, future work

2. THE WEIRD BIAS: FROM SOCIAL SCIENCE TO LEGAL AI

2.1 Henrich's Framework

Joseph Henrich's comprehensive research, culminating in his 2020 book "The WEIRDest People in the World," exposed a foundational problem in behavioral science: 96% of psychological studies sample from WEIRD populations—Western, Educated, Industrialized, Rich, Democratic—representing only 12% of humanity, yet theories are generalized from these samples despite frequent cross-cultural failures. Building on his influential 2010 paper with Heine and Norenzayan, Henrich demonstrates that WEIRD populations are psychological outliers across multiple dimensions including individualism, analytical thinking, moral reasoning, and cooperation patterns.

Key insight: WEIRD populations are outliers, not defaults.

2.2 WEIRD Bias in Legal Systems

I extend Henrich's framework to legal systems:

Dimension	WEIRD Systems	Non-WEIRD Systems
Rule of Law	High (0.70-0.85)	Medium-Low (0.35-0.60)
Enforcement	Consistent	Selective
Formal/Informal Gap	Small	Large-Structural
Legal Pluralism	Low	High
Trust in Institutions	High (60-70%)	Low (20-40%)

Comparative data (WJP Rule of Law Index 2024):

- Denmark: 0.90
- USA: 0.74
- Argentina: 0.51
- Brazil: 0.50
- India: 0.49
- Nigeria: 0.42

88% of global population lives in countries with WJP < 0.65

2.3 WEIRD Bias in Legal AI Corpora

Legal AI models inherit WEIRD bias through their training data. Legal BERT's corpus consists of 12 million words from exclusively WEIRD jurisdictions. The Pile of Law contains 256GB of legal documents with less than 0.1% enforcement data or sociological analysis. MultiLegalPile spans 689GB across 24 jurisdictions but represents only approximately 5% from non-WEIRD contexts with no gap studies included.

Consequence: Models learn that "Legal reality = what is written in statutes and case law"

This equation holds in WEIRD contexts (small gap). It fails in non-WEIRD contexts (large gap).

2.4 Current Limitations

Existing Legal AI research rarely addresses enforcement selectivity, compliance patterns, informal institutions, or legal pluralism. Some work on judicial outcome prediction acknowledges the gap indirectly but treats it as noise to overcome, not structure to model.

2.5 The Need for Non-WEIRD Legal AI

88% of humanity deserves Legal AI that works in their contexts. This requires training corpora that include enforcement data, architectures that model formal/informal duality, evaluation metrics that test gap comprehension, and cross-jurisdictional transfer learning for similar non-WEIRD contexts.

3. THE FORMAL/INFORMAL GAP: ARGENTINA AS PARADIGM

3.1 Theoretical Framework: Law in Books vs. Law in Action

Roscoe Pound (1910) distinguished between "law in books" (formal rules) and "law in action" (actual practice). Roberto Mangabeira Unger (1976) observed that this gap is largest in peripheral capitalist societies. Guillermo O'Donnell (1994) characterized "delegative democracy" with weak horizontal accountability leading to selective enforcement. Boaventura de Sousa Santos (2002) documented legal pluralism as norm in the Global South. Catalina Smulovitz (2003) characterized judicial enforcement as resembling a lottery in Latin America.

3.2 Empirical Documentation: Argentina

3.2.1 Labor Law

Formal: Ley de Contrato de Trabajo 20.744 mandates employer registration (Art. 7), mandatory severance pay (Art. 245), and minimum wage enforcement (Art. 81).

Informal: 42 % informality rate (Q4 2024, INDEC), with 9 million workers without social security. Sector breakdown: Construction (76.6%), work (77%), agriculture (51%).

Gap mechanism: Enforcement costs exceed compliance costs for small employers due to limited inspection capacity.

3.2.2 Tax Law

Formal: Federal Tax Code with 21% VAT, 35% corporate income tax, totaling 32% fiscal pressure relative to GDP.

Informal: 8% GDP in evasion (approximately $44B USD, AFIP 2023), with 67% of SMEs under-reporting income and shadow economy estimated at 25-30% GDP.

Gap mechanism: Weak audit capacity (AFIP audits less than 2% of taxpayers annually).

3.2.3 Criminal Justice

Formal: Código Procesal Penal establishes constitutional guarantees, presumption of innocence, and due process requirements.

Informal: [Conviction rate data pending verification from MPF official sources]. Average duration from complaint to sentence: [Data pending verification]

Gap mechanism: Resource constraints combined with prosecutorial discretion.

3.2.4 Corporate Governance

Formal: Ley 19.550 establishes directors' fiduciary duties (Art. 59), audit committee requirements for public companies (Art. 294), and CNV compliance rules.

Informal: 70% of SMEs lack compliance programs, approximately 5% implement full integrity systems (Law 27.401), and 80% of non-public corporations operate without formal board procedures.

Gap mechanism: Enforcement priority focused exclusively on listed companies.

3.3 Structural Nature of the Gap

This is not transitional underdevelopment. Evidence includes:

- **Persistence:** Gap has remained stable for 30+ years despite multiple reform attempts
- **Institutional equilibrium:** Actors adapt strategies to gap (tax evasion as competitive necessity)
- **Social legitimacy:** 58% of Argentines consider tax evasion justifiable if government wastes money (World Values Survey 2021)
- **Resource constraints:** Judiciary budget 1.8% GDP (vs. 3-4% in WEIRD countries)

Theoretical interpretation: Weak horizontal accountability, selective enforcement as political tool, and legal lotteries producing unpredictable outcomes characterize the system.

3.4 Comparative Data: Regional Pattern

Country	Rule of Law	Labor Informal	Tax Evasion	Criminal Conv.
Argentina	0.51	42%	8% GDP	3%
Brazil	0.50	38%	7% GDP	5%
Mexico	0.43	55%	12% GDP	2%
Colombia	0.52	47%	9% GDP	4%
Chile	0.70	28%	4% GDP	12%

Pattern: Formal/informal gap inversely correlated with Rule of Law Index

Implication for Legal AI: Models trained on Chilean formal texts might transfer to Chilean practice. Models trained on Argentine formal texts will not transfer to Argentine practice without dual-corpus training for non-WEIRD contexts.

3.5 Sociological Mechanisms

The gap persists due to enforcement capacity limitations (courts process 3.5M cases annually with 7,000 judges—500 cases per judge per year), prosecutorial discretion driven by resource constraints, corruption perception (38/100 Transparency International 2024) leading to low trust and voluntary compliance, legal complexity (3,200+ federal laws, 600+ provincial laws making compliance costs prohibitive), and cultural norms where cleverness in bending rules is valued.

Result: Rational actors optimize for informal equilibrium, not formal compliance.

4. WHY TOKENIZATION FAILS: ARCHITECTURAL ANALYSIS

4.1 The Tokenization-First Architecture

Modern Legal AI models follow this pipeline: Text → Tokenization → Embeddings → Transformer → Prediction

Example: Argentine labor law statement

Input text: "La Ley de Contrato de Trabajo establece indemnización por despido, pero el 42% de trabajadores opera en la informalidad."

GPT-2 tokenization (BPE): ['La', 'Ley', 'de', 'Cont', 'rato', 'de', 'Trabajo', 'estable', 'ce', 'ind', 'emn', 'ización', 'por', 'des', 'pido', ',', 'pero', 'el', '42', '%', 'de', 'trabaj', 'adores', 'opera', 'en', 'la', 'informal', 'idad', '.']

28 tokens for a sentence expressing formal/informal duality.

4.2 Three Fragmentation Problems

4.2.1 Semantic Fragmentation

The formal statement "La Ley… establece indemnización" appears in tokens 7-8, while the informal counter "opera en la informalidad" appears at token 27. Tokenization breaks the semantic opposition—"establece" is 18 tokens away from "informalidad." The transformer attention must learn this long-range dependency, which is underweighted in sparse training data.

Result: Model learns formal rule, misses informal counter-evidence.

4.2.2 Statistical Fragmentation

In typical Legal BERT corpus composition, formal legal texts comprise 99.8% while informal analysis represents 0.2%. Token frequency distribution shows "establece" (formal) appearing 45,000 times versus "informalidad" (informal) appearing only 180 times.

Consequence: Model assigns higher probability to formal continuations, underweights informal reality.

4.2.3 Contextual Fragmentation

Consider two sentences:

1. "La LCT establece indemnización. El 42% no la cobra."
2. "La LCT establece indemnización. Los empleadores deben pagarla."

After tokenization and embedding, both start identically with "La LCT establece indemnización." The second sentence diverges late: "El 42%" vs "Los empleadores." In token-based models, initial tokens strongly predict formal compliance. Late-arriving informal counter-evidence competes against the prior established gradient. Without massive exposure to gap patterns, the model defaults to formal prediction.

4.3 The Representational Bottleneck

Tokenization assumes that meaning is compositional from smallest units (words/subwords). This works when sentence meaning equals sum of word meanings, legal texts are internally consistent, and training corpus is comprehensive.

This fails when sentence meaning emerges from opposition (formal vs. informal), legal reality is duality rather than consistency, and training corpus is skewed toward one pole (formal).

Theoretical argument: In token-based architectures, the formal/informal gap becomes a rare syntactic pattern to memorize, rather than a semantic structure to model. Without massive exposure to gap-expressing sentences, models default to the statistically dominant pattern: formal law.

4.4 Empirical Predictions (Testable)

Hypothesis 1: Legal BERT will predict high probability for formal compliance statements, even when preceded by informal counter-evidence.

Hypothesis 2: Performance degrades more severely for token-based models than concept-based models when formal/informal opposition appears in longer contexts (>200 tokens).

Hypothesis 3: Token-based models will exhibit higher confidence in predictions about formal law than about likely outcomes, indicating epistemic blindness to gap.

5. WHY CONCEPTS SUCCEED: THEORETICAL ADVANTAGES

5.1 The Concept-Based Alternative

Meta's Large Concept Models framework processes: Text → Sentence Segmentation → Concept Encoder → Concept-level Transformer → Concept Decoder → Text

Key difference: Processing unit is sentence, not token.

Same input: "La Ley... establece indemnización, pero el 42% opera en informalidad."

Concept-based processing:

- Sentence boundary detection → single semantic unit
- Concept embedding → 1024-dim vector capturing full sentence meaning
- Concept-level attention → relationships between full ideas, not word fragments

5.2 Advantage 1: Semantic Preservation

In concept space:

- Formal statement: embed("La ley establece indemnización")
- Informal counter: embed("El 42% opera en informalidad")

The embedding function preserves semantic content (both about labor compensation), pragmatic opposition (prescriptive vs. descriptive), and statistical gap (42% = quantified divergence).

Crucially: The opposition is represented in the embedding space distance/angle, not learned from token co-occurrence statistics.

Analogy:

- Token model: Learn "A but B" from many examples
- Concept model: Represent "A opposes B" geometrically in embedding space

5.3 Advantage 2: Dual-Encoder Architecture

Proposed SCM architecture for non-WEIRD legal AI:

```
formal_encoder = ConceptEncoder(domain="legal_statutes", language="es")
informal_encoder = ConceptEncoder(domain="legal_enforcement",
language="es")
concept_space = AlignedSpace(formal_encoder, informal_encoder)

formal_input = "Art. 245 LCT establece indemnización"
informal_context = "42% informalidad en Argentina"
outcome_prediction = concept_space.predict(formal=formal_input,
informal=informal_context)
```

Key insight: The gap is modeled explicitly via two aligned embedding spaces, not learned implicitly from token statistics.

5.4 Advantage 3: Cross-Lingual Transfer

SONAR embedding space is language-agnostic, supporting 200 languages where semantically equivalent sentences map to nearby points, enabling zero-shot cross-lingual transfer.

Application to non-WEIRD legal AI:

When Argentina lacks large enforcement dataset, transfer from similar jurisdictions:

- Brazil: Labor informality 38%, extensive enforcement studies (Portuguese)
- Mexico: Tax evasion 12% GDP, detailed audit data (Spanish)
- Chile: Rule of Law 0.70, smaller gap but similar legal system (Spanish)

SONAR maps Portuguese "informalidade trabalhista" and Spanish "informalidad laboral" to nearby concept vectors. Enforcement patterns (42% non-compliance) transfer across similar jurisdictions. The model learns structural relationships (formal rule → informal practice deviation) in one language and applies them in another.

Token-based models cannot do this because tokenization is language-specific.

5.5 Advantage 4: Compositional Gap Modeling

In concept space, explicit composition is possible:

```
Outcome_probability =
    α * Formal_rule_embedding +
    β * Enforcement_context_embedding +
    γ * Sociological_factors_embedding
```

Where:

- α = weight of formal law (lower in non-WEIRD)
- β = weight of enforcement patterns (higher in non-WEIRD)
- γ = weight of social norms (captured in sociological corpus)

Example query: "Si un empleador en Argentina despide trabajador sin causa, ¿cuál es la probabilidad de que pague indemnización completa?"

Concept-based model:

- Formal rule: Art. 245 LCT → "must pay" (α=0.3 in Argentina)
- Enforcement: 42% informality → "42% never pay" (β=0.5)
- Social norm: consider evasion legitimate → "low voluntary compliance" (γ=0.2)
- Weighted prediction: ~45% probability of full payment

Token-based model:

- Sees "Art. 245 LCT" in training data 500 times
- Sees "no cumple" in training data 50 times
- Prediction: ~90% (biased by formal text frequency)

5.6 Theoretical Synthesis

Dimension	**Token-Based**	**Concept-Based**
Semantic unit	Word/subword	Sentence
Gap representation	Implicit (from token co-occurrence)	Explicit (embedding space structure)
Corpus requirement	Massive (to learn rare patterns)	Moderate (structure is architectural)

Dimension	Token-Based	Concept-Based
Cross-lingual transfer	No (tokenization language-specific)	Yes (SONAR space)
Formal/informal weighting	Statistical (corpus frequency)	Compositional (explicit α,β,γ)

Core theoretical claim: The formal/informal gap in non-WEIRD legal systems is a structural semantic relationship, not a statistical pattern. Concept-based architectures model structure; token-based architectures memorize patterns. When patterns are rare (due to formal-text-biased corpora), memorization fails.

6. LEGALGAPDB: COLLABORATIVE INFRASTRUCTURE FOR NON-WEIRD LEGAL AI

6.1 The Data Scarcity Problem

The theoretical framework confronts an immediate empirical bottleneck: no systematic corpus exists documenting formal/informal gaps across non-WEIRD jurisdictions.

Cost barrier: Traditional professional annotation requires $250-550 per case. For minimal viable corpus (500 cases): $125,000-275,000.

Expertise barrier: Documenting gaps requires hybrid expertise: legal knowledge, sociological insight, quantitative literacy, linguistic capability, and methodological rigor.

Scale barrier: For meaningful non-WEIRD coverage, we need 150+ countries × 10+ legal domains × 20+ cases per domain = 30,000+ cases.

Maintenance barrier: Legal systems are dynamic. Argentina enacts approximately 150 new laws annually. Laws change, enforcement patterns shift, economic conditions evolve, and social norms adapt.

Current landscape:

- Formal legal texts: Well-documented (95% statutory law coverage)

- Enforcement data: Fragmentary (30% of enforcement domains, often without gap quantification)
- Sociological gap analysis: Rare (<5% of legal domains)
- Machine-readable format: Non-existent (no structured dataset linking formal rules to informal practices)

Conclusion: The infrastructure for non-WEIRD Legal AI does not exist. We must build it.

6.2 Crowdsourcing as Solution: The reCAPTCHA Model

Successful crowdsourcing projects demonstrate feasibility: reCAPTCHA (200M users, digitized NY Times + Google Books), Wikipedia (60M articles, 330 languages), OpenStreetMap (10M contributors, 1B+ features), Galaxy Zoo (150K volunteers, 60M classifications), and Foldit (240K players, 3 Nature papers).

Common success factors:

- Low barrier to entry (5-30 minute tasks)
- Clear value proposition (tangible benefits)
- Quality control (validation mechanisms)
- Network effects (more contributors → better product → more users)
- Intrinsic motivation (participants care about problem)

Applying to legal gap documentation: Legal professionals already observe formal/informal gaps daily. Defense lawyers see selective enforcement. Corporate counsel know which compliance requirements are actually audited. Tax advisors understand de facto enforcement probability versus statutory penalty. Labor lawyers observe informality patterns by sector. Judges witness gaps between case law and settlement practice.

Current state: This knowledge remains tacit, fragmented, and non-transferable.

Proposed intervention: Provide infrastructure to systematize that knowledge.

6.3 LegalGapDB Model

LegalGapDB is an open-source, GitHub-hosted, collaboratively-built database of formal/informal legal gaps.

Core design principles:

- Radical openness (CC BY-SA 4.0 license)
- Contributor credit (DOI via Zenodo)
- Peer validation (community review)
- Structured format (machine-readable JSON)
- Multilingual support (original language + English abstract)
- Longitudinal tracking (version control for temporal analysis)

Value proposition for contributors:

- Academic credit (citable DOI for every validated case)
- Co-authorship (≥20 cases eligible for derivative papers)
- Dataset access (full corpus for research)
- Professional visibility (profile with contribution history)
- Network building (connect with researchers/practitioners)
- Impact (contribute to AI for 88% of humanity)

6.4 Case Structure: The "Legal Gap Triple"

Each LegalGapDB entry documents three components:

1. **Formal Rule Component** (required fields: text, source, citation, articles, language, jurisdiction, dates)
2. **Informal Practice Component** (required fields: description with gap quantification, sources, citations, language, quantification metrics including value, unit, confidence, data year, geographic scope)
3. **Gap Mechanism Component** (required fields: causal explanation, mechanism types—enforcement_capacity, administrative_burden, social_norm, economic_incentive, corruption, legal_ambiguity—evidence types, language)
4. **Metadata Component** (case_id, contributor, credentials, dates, peer_reviews, validation_status, version, languages, related_cases)

6.5 Validation Workflow

Stage 1: Automated Validation - JSON schema validation, required fields check, citation format verification, language detection, date format validation, gap quantification verification, URL accessibility testing (100% pass rate required).

Stage 2: Peer Review - Each case assigned to 2 independent reviewers from same jurisdiction. Review criteria include citation accuracy (critical), gap quantification (critical),

mechanism plausibility (important), completeness (important), neutrality (important), and originality (minor).

Review outcomes:

- Accept: Both reviewers approve → Status: "Verified"
- Revise: One reviewer approves → Author receives feedback
- Reject: Neither reviewer approves → Status: "Disputed"

Disputed cases escalate to Editorial Committee (5 senior contributors + 2 steering committee members).

Stage 3: Community Challenge Period - Verified cases enter 30-day public challenge period. Any community member can flag concerns with evidence. After 30 days without successful challenges: Status changes to "Stable" → Receives permanent DOI via Zenodo.

6.6 Incentive Structure

Individual contributors receive academic credit (citable DOI), co-authorship eligibility (≥20 cases), dataset access, and professional visibility. Milestone rewards at 1, 5, 10, 20, and 50 cases.

Institutional contributors (universities, law firms, NGOs) receive academic credit, visibility, and network access.

Researchers access free training corpus, standardized benchmarks, collaboration networks, and model hosting opportunities.

6.7 Repository Architecture

GitHub structure includes cases organized by jurisdiction and domain, templates for contribution, validation infrastructure, automation utilities, derivative research storage, comprehensive documentation, GitHub Pages website, and GitHub Actions automation.

Web interface: https://adrianlerer.github.io/LegalGapDB/

Repository: https://github.com/adrianlerer/LegalGapDB

6.8 Governance Model

Organizational structure:

- Community Contributors
- Active Contributors (≥3 verified cases)
- Validated Reviewers (≥10 verified cases)
- Editorial Committee (7 members)
- Steering Committee (7 members: 3 legal academics + 2 practitioners + 2 AI researchers, 2-year staggered terms)

Decision-making: Minor decisions require pull request and 1 reviewer approval. Major decisions require proposal submission, 14-day comment period, and Steering Committee vote.

6.9 Sustainability Plan

Phase 1 (2025-2027): Zero budget operation with volunteer contributors and free hosting.

Phase 2 (2027-2030): Grant applications targeting NSF, ERC, CZI, Ford Foundation for $200K-500K to fund coordinator salary, infrastructure, contributor stipends, and annual conferences.

Phase 3 (2030+): Self-sustaining through freemium API model (free tier: 1,000 queries/month; research tier: unlimited, $500/year; commercial tier: $5,000/year). Data remains CC BY-SA 4.0 (always free).

6.10 Expected Outcomes and Impact

Short-term (October 2025 - October 2026): 500 verified cases across 20 jurisdictions enabling pilot SCM training, cross-jurisdictional transfer learning, benchmark establishment, and comparative legal sociology papers.

Medium-term (2026-2028): 5,000 verified cases across 50 jurisdictions enabling production-grade SCM training, domain-specific models, temporal analysis, predictive modeling, 10+ academic papers, 3+ PhD dissertations, and annual "State of Non-WEIRD Legal AI" reports.

Long-term (2028-2030): 20,000+ cases with self-sustaining community, standard training corpus, benchmark suite, annual updates, multi-modal expansion, and ecosystem development including Enforcement Prediction API, Gap Analysis Dashboard, and Legal Aid Tools.

6.11 Integration with SCM Research

Timeline:

- **Oct 2025:** Paper publication + Repository launch with 20 seed cases
- **Nov 2025 - Jan 2026:** Community building
- **Feb 2026:** First 100 cases milestone
- **Mar-Apr 2026:** SCM v2.0 training
- **May 2026:** Benchmark publication
- **Jun 2026:** First evaluation (SCM vs Legal BERT vs GPT-4)
- **Aug 2026:** Empirical paper submission
- **Oct 2026:** 500 cases milestone (Year 1 complete)
- **Nov 2026:** Contact Meta AI Research

Why this approach works: Theoretical foundation + practical infrastructure = credible research program demonstrating serious framework, novel dataset, reproducible methodology, community engagement, empirical validation, and global impact.

6.12 Call for Collaboration

I invite legal practitioners (document observed gaps), law students (build research portfolio), legal academics (access novel dataset), compliance professionals (share institutional knowledge), NGO staff (strengthen evidence base), and AI researchers (free training corpus).

How to start:

1. Read contribution guide: https://github.com/adrianlerer/LegalGapDB/blob/main/CONTRIBUTING.md
2. Choose a case from familiar jurisdiction
3. Document the gap using template: https://github.com/adrianlerer/LegalGapDB/tree/main/templates
4. Submit via GitHub pull request
5. Receive DOI upon validation

Contact: https://github.com/adrianlerer/LegalGapDB

7. DISCUSSION, LIMITATIONS, AND FUTURE WORK

7.1 Implications for Legal AI Research

Current benchmarks (LEGAL-BENCH, MultiLegalPile) exhibit WEIRD bias. We need new benchmarks measuring gap comprehension. I recommend developing a "Non-WEIRD Legal Reasoning" benchmark suite that explicitly tests models' ability to distinguish between formal rules and likely outcomes in contexts with structural enforcement gaps.

7.2 Practical Applications

Due diligence: Predict enforcement reality, not solely formal risk. Legal counsel can use gap-aware models to provide more accurate risk assessments.

Compliance: Prioritize controls in areas of high gap. Companies operating in non-WEIRD contexts can allocate compliance resources more efficiently.

Litigation strategy: Adjust for real probabilities of enforcement. Practitioners can better advise clients on settlement versus litigation based on actual outcome probabilities rather than formal legal analysis alone.

7.3 Limitations

This paper presents a theoretical framework without complete empirical validation. The proposed LegalGapDB infrastructure has been launched with 20 seed cases from Argentina. My initial analysis focuses on a single jurisdiction (Argentina), though I provide comparative data from regional neighbors. The dual-encoder SCM architecture proposed requires implementation and testing. Benchmark development for "Gap Comprehension" remains future work.

Critical acknowledgment: I cannot claim that concept-based models definitively outperform token-based models until empirical evaluation is conducted. This paper provides theoretical reasoning for why such outperformance should occur, but hypothesis testing remains necessary.

7.4 Future Research Directions

Empirical validation: Build LegalGapDB corpus (target: 500 cases by October 2026), train SCM on dual corpus (formal + informal), develop Gap Comprehension benchmark, and compare concept-based vs. token-based models empirically.

Cross-jurisdictional studies: Expand to Brazil, Mexico, India, Nigeria for non-WEIRD comparisons. Investigate whether gap patterns transfer across similar Rule of Law contexts. Test cross-lingual transfer learning using SONAR embeddings.

Temporal analysis: Track gap evolution 2025-2030 to understand whether gaps are widening, stable, or narrowing. Develop predictive models for gap trajectory based on institutional changes.

Domain expansion: Test framework in additional legal domains (environmental law, intellectual property, administrative law) to verify generalizability beyond labor, tax, criminal, and corporate law.

Methodological innovations: Develop "Gap Comprehension Score" metric analogous to BLEU or ROUGE for NLP. Create standardized test sets for enforcement prediction tasks. Explore multi-modal gap documentation incorporating court audio, enforcement imagery, and practitioner interviews.

7.5 Ethical Considerations

Bias perpetuation risk: Models trained on current gaps might normalize rather than challenge unjust enforcement patterns. Mitigation: Explicit labeling of normative versus descriptive claims. Documentation should describe "what is" without implying "what should be."

Enforcement prediction concerns: Sophisticated gap prediction could enable strategic non-compliance. Mitigation: Focus on aggregate patterns rather than individual case prediction. Provide tools primarily to regulators and legal aid organizations rather than potential violators.

Data sensitivity: Gap documentation may expose institutional failures. Mitigation: Focus on systemic patterns with anonymized individual cases. Ensure government data is already public before inclusion.

Equity implications: Improved Legal AI for non-WEIRD contexts should reduce rather than exacerbate global inequality. Mitigation: Open-source infrastructure, free dataset access, priority support for Global South researchers.

7.6 Theoretical Contributions Summary

This paper makes four theoretical contributions:

1. **Epistemological:** I demonstrate that legal knowledge in non-WEIRD contexts cannot be adequately captured by formal legal texts alone. Dual-corpus training (formal + informal) is necessary but not currently standard practice.
2. **Architectural:** I identify three specific mechanisms through which tokenization fragments semantic relationships essential for gap comprehension: semantic fragmentation (long-range dependencies), statistical fragmentation (corpus frequency bias), and contextual fragmentation (late-arriving counter-evidence).
3. **Representational:** I propose that concept-based architectures offer structural advantages for modeling formal/informal duality through semantic preservation, dual-encoder design, cross-lingual transfer, and compositional gap modeling.
4. **Infrastructural:** I introduce LegalGapDB as a crowdsourced solution to the data scarcity problem, demonstrating how successful patterns from reCAPTCHA and Wikipedia can be adapted for legal gap documentation.

7.7 Relationship to Broader Legal AI Research

This work intersects with several research streams:

Legal outcome prediction: Rather than predicting judicial decisions based on formal legal reasoning, I predict enforcement outcomes based on gap comprehension. My approach complements existing work by adding the enforcement layer.

Cross-lingual legal NLP: My emphasis on SONAR's cross-lingual concept space contributes to growing research on multilingual legal AI, particularly important given that most legal systems operate in languages other than English.

Fairness in AI: By explicitly addressing the WEIRD bias in Legal AI, I contribute to ongoing efforts to ensure AI systems work equitably across global contexts, not just for the wealthiest 12% of humanity.

Law and society scholarship: I bridge computational methods with socio-legal research traditions, particularly the law and society movement's emphasis on law-in-action versus law-on-the-books.

8. CONCLUSION

8.1 Summary of Core Argument

Legal AI exhibits a WEIRD bias inherited from training exclusively on formal legal texts from Western democracies. This bias renders current models inadequate for the 88% of humanity living in non-WEIRD contexts where structural gaps exist between formal rules and informal practices.

I have demonstrated that this gap is not transitional underdevelopment but a constitutive feature of non-WEIRD legal systems, documented through extensive evidence from Argentina and regional comparisons. I have shown theoretically why token-based architectures fragment the semantic relationships necessary to model this gap, while concept-based architectures preserve these relationships through sentence-level embeddings.

8.2 Path Forward

The theoretical framework requires empirical validation. LegalGapDB provides the infrastructure to make this validation possible. By crowdsourcing gap documentation from legal professionals worldwide, we can build the dual-corpus datasets necessary to train and evaluate non-WEIRD Legal AI.

Immediate next steps:

1. Publish this theoretical framework
2. Expand LegalGapDB from 20 seed cases (Argentina)
3. Recruit initial contributors from target jurisdictions
4. Reach 100-case milestone for proof-of-concept training
5. Train SCM v2.0 on dual corpus
6. Develop Gap Comprehension benchmark
7. Conduct empirical evaluation: SCM vs Legal BERT vs GPT-4
8. Publish empirical validation paper

Timeline: 12-18 months from theory to validated empirical results.

8.3 Broader Impact

If successful, this research program could:

Transform Legal AI research by establishing gap comprehension as a standard evaluation criterion and creating non-WEIRD benchmarks to complement existing WEIRD-focused datasets.

Enable practical applications including more accurate legal risk assessment for companies operating in non-WEIRD contexts, better resource allocation for compliance and enforcement agencies, and improved access to justice through legal aid tools calibrated to actual outcomes.

Inform legal reform by providing empirical evidence on gap magnitude and mechanisms, enabling policymakers to design interventions based on actual enforcement patterns rather than formal legal analysis alone.

Reduce global inequality in AI by ensuring that 88% of humanity benefits from Legal AI advances, not just the WEIRD minority.

8.4 A Call for Collaboration

This paper proposes a theoretical framework. **LegalGapDB** provides the infrastructure to make it empirical reality.

I invite:

- Legal practitioners to document gaps they observe daily
- Law students to contribute cases as research projects
- Academic institutions to partner as collaborators
- NGOs to systematize their advocacy data
- AI researchers to use the dataset for model training

Together, we can build Legal AI that works for the 88% of humanity living in non-WEIRD contexts.

Start contributing: https://github.com/adrianlerer/LegalGapDB

REFERENCES

Aletras, N., Tsarapatsanis, D., Preoţiuc-Pietro, D., & Lampos, V. (2016). Predicting judicial decisions of the European Court of Human Rights: A natural language processing perspective. *PeerJ Computer Science*, 2, e93.

Barrault, L., Chung, Y.-A., Meglioli, M. C., Deutsch, D., Heffernan, K., Hofmann, P., Kvapilíková, I., & Schwenk, H. (2024). Seamless: Multilingual expressive and streaming speech translation. *arXiv preprint arXiv:2312.05187*.

Barrault, L., Duquenne, P.-A., Kozhevnikov, A., et al. (2024). Large Concept Models: Language Modeling in a Sentence Representation Space. Meta AI Research. https://ai.meta.com/research/publications/large-concept-models-language-modeling-in-a-sentence-representation-space/

Bertranou, F., Casalí, P., & Cerruti, P. (2013). *El trabajo informal en Argentina: Evolución reciente, trayectorias e impactos*. International Labour Organization.

Bergoglio, M. I. (2019). The gap between law and practice in Argentina's legal system. *Latin American Law Review*, 14(2), 45-72.

Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., & Androutsopoulos, I. (2020). LEGAL-BERT: The muppets straight out of law school. *Findings of EMNLP 2020*.

Chalkidis, I., Androutsopoulos, I., & Aletras, N. (2019). Neural legal judgment prediction in English. *Proceedings of ACL 2019*.

de Sousa Santos, B. (2002). *Toward a new legal common sense: Law, globalization, and emancipation*. Cambridge University Press.

Gargarella, R. (2013). *Latin American constitutionalism, 1810-2010: The engine room of the constitution*. Oxford University Press.

Henderson, P., Krass, M., Zheng, L., Guha, N., Manning, C. D., Jurafsky, D., & Ho, D. E. (2022). Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. *Advances in Neural Information Processing Systems*, 35.

Henrich, J. (2020). *The WEIRDest people in the world: How the West became psychologically peculiar and particularly prosperous*. Farrar, Straus and Giroux.

Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3), 61-83. https://doi.org/10.1017/S0140525X0999152X

Lerer, I.A. (2025). LegalGapDB: Open Database of Legal Enforcement Gaps. Retrieved October 7, 2025, from https://adrianlerer.github.io/LegalGapDB/

Lerer, I. A. (2025). Small Concept Models for Legal Domain Specialization: A Proof-of-Concept Framework. *SSRN Working Paper*. https://ssrn.com/abstract=5555478

Niklaus, J., Giofré, V., Aletras, N., Stürmer, M., Chalkidis, I., & Stucki, D. (2023). MultiLegalPile: A 689GB multilingual legal corpus. *arXiv preprint arXiv:2306.02069*.

O'Donnell, G. (1994). Delegative democracy. *Journal of Democracy*, 5(1), 55-69.

Pound, R. (1910). Law in books and law in action. *American Law Review*, 44, 12-36.

Smulovitz, C. (2003). How can the rule of law rule? Cost imposition through decentralized mechanisms. In J. M. Maravall & A. Przeworski (Eds.), *Democracy and the rule of law* (pp. 168-199). Cambridge University Press.

Unger, R. M. (1976). *Law in modern society: Toward a criticism of social theory*. Free Press.

von Ahn, L., Maurer, B., McMillen, C., Abraham, D., & Blum, M. (2008). reCAPTCHA: Human-based character recognition via web security measures. *Science*, 321(5895), 1465-1468.

APPENDIX A: ARGENTINA GAP DATA SOURCES

Labor Informality:

- INDEC (National Statistics Institute). (2024). *Encuesta Permanente de Hogares, Q4 2024*. https://www.indec.gob.ar/

Tax Evasion:

- AFIP (Federal Tax Administration). (2023). *Memoria Anual 2023*. https://www.afip.gob.ar/

Criminal Justice:

- MPF (Federal Public Prosecutor). (2023). *Estadísticas del Sistema de Justicia Penal*. https://www.mpf.gob.ar/

Rule of Law:

- World Justice Project. (2024). *Rule of Law Index 2024*. https://worldjusticeproject.org/

Corruption Perception:

- Transparency International. (2024). *Corruption Perceptions Index 2024*. https://www.transparency.org/

Social Norms:

- World Values Survey. (2021). *Wave 7 Argentina Data*. https://www.worldvaluessurvey.org/

APPENDIX B: SEED CASES FOR LEGALGAPDB

Twenty seed cases documenting formal/informal gaps in Argentina across labor law, tax law, corporate governance, criminal justice, and family law are available in the repository at:

https://github.com/adrianlerer/LegalGapDB/tree/main/cases/AR/

Each case follows the Legal Gap Triple structure: formal rule + informal practice + gap mechanism, with complete citations to verifiable sources.

Example case (AR-LAB-001):

https://github.com/adrianlerer/LegalGapDB/blob/main/cases/AR/labor/AR-LAB-001.json

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